

TransitIQ

Detailed Project Report

AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet

Smart India Hackathon 2026

Problem Statement ID: SIH26124

Team #ArtisanX · Bhopal Pilot Corridor

This report documents the TransitIQ prototype, its proposed system workflow, technical architecture, dashboard capabilities, feasibility, impact, and prototype scope.

Prototype note: The current dashboard is a demonstration prototype. Bhopal locations, bus routes, GPS points, road measurements and individual detections shown in the demo are simulated data rather than surveyed municipal data.

1. Project Overview

TransitIQ is a smart-city mobility concept that turns public buses into moving road-health sensors. Road-facing cameras and GPS units are proposed to collect road-condition observations during normal bus journeys. The system then geo-tags findings, prioritizes them, routes them to the relevant city department and tracks resolution through a central dashboard.

Problem Addressed

Road-condition and street-infrastructure problems can be difficult to monitor continuously. TransitIQ uses the existing public-transport network as a distributed sensing layer so that regular bus routes can repeatedly observe the same road network.

Prototype Goals

- Visualize a live public-bus fleet across a Bhopal pilot corridor.
- Represent AI-detected road and street-infrastructure issues.
- Geo-tag detections and display them on a city map.
- Score and prioritize issues for response.
- Route issues to relevant departments and track their status.
- Provide road-health, route, analytics, alert and resolution views.

Issue Categories in the Prototype

Issue	Prototype department
Pothole	Road Maintenance Dept
Garbage	Sanitation Dept
Broken Streetlight	Electrical Dept
Faded Road Marking	Traffic & Roads Dept
Damaged Road Sign	Traffic Signage Unit

2. Methodology & Process of Implementation

1. Capture

Road-facing camera and GPS on the bus provide continuous road observations and location.

2. Detect

The proposed architecture uses Edge AI / YOLOv8 to identify road and street-infrastructure issues.

3. Geo-tag

Each detection is associated with coordinates and a timestamp.

4. Sync

MQTT over 4G/5G is proposed for lightweight transmission when connectivity is available.

5. Store & Analyze

A FastAPI backend and PostGIS database are proposed for ingestion, spatial queries and analytics.

6. Prioritize

The platform scores issues and highlights higher-priority findings for city teams.

7. Assign

Issues are routed to the relevant department for action.

8. Resolve

Resolution status and response time are tracked; the dashboard provides a history of the workflow.

Technical Architecture

Layer	Proposed technology / role
Sensing	Bus camera + GPS
Edge intelligence	Edge AI / YOLOv8
Connectivity	MQTT over 4G / 5G
Backend	Python / FastAPI
Spatial database	PostGIS
Analytics	Priority scoring, trends and department routing
Dashboard	React + Leaflet / OpenStreetMap

Prototype implementation: The supplied prototype is a self-contained browser dashboard using React 18, ReactDOM, Recharts, Lucide React and Leaflet loaded through browser CDNs. The current demo simulates the fleet, detections, routes and city data so the workflow can be demonstrated without a production backend.

3. Prototype Modules

Dashboard — High-level command-center view of fleet activity, detections, priorities and recent agent activity.

Live Fleet — Map and roster for simulated buses, including route, speed, GPS, camera and network state.

AI Detections — Detection list for potholes, garbage, broken streetlights, faded markings and damaged signs.

City Map — Map-based view of buses and issue markers across the Bhopal pilot corridor.

Road Health — Road-health scoring and category views; includes a monsoon-mode toggle and re-scan freshness indicators.

Smart Routes — Route-focused view intended to support road-aware routing and route comparison.

Analytics — Charts and summary information for trends and operational monitoring.

Alerts — Operational alerts for issues requiring attention.

Resolution Center — Department assignment, status tracking and resolution history.

Settings — Prototype configuration and display controls.

Bhopal Pilot Representation

The prototype uses recognizable Bhopal areas such as MP Nagar, TT Nagar/New Market, Habibganj, BHEL, Karond, Ayodhya Bypass, Bittan Market, Kolar Road, Bairagarh, Chunabhatti and Piplani. The prototype explicitly treats the road widths, lane counts, GPS points, bus routes and individual detections as simulated demonstration data.

Example simulated fleet

The dashboard contains simulated buses including BUS-104, BUS-217, BUS-305, BUS-411, BUS-508, BUS-612, BUS-733, BUS-820, BUS-905 and BUS-1002, with route, speed, GPS, camera and connectivity states.

4. Feasibility, Viability & Impact

Feasibility

- Reuses public buses that already travel through the city.
- Uses commonly available sensing, connectivity and open-source software components.
- Edge processing can reduce the need to transmit continuous raw video.
- A pilot corridor can be deployed before city-wide expansion.

Key Challenges & Mitigation Direction

Challenge	Mitigation direction
Blur / changing light	Stable camera mounting and training/validation on varied conditions.
Weak connectivity	Transmit compact detections and synchronize when connectivity returns.
Integration with city systems	Use APIs and begin with a controlled pilot.
False detections	Confidence thresholds, verification workflow and feedback from resolved cases.

Expected Benefits

- Safer roads through earlier identification of hazards.
- Faster response by routing findings to the appropriate city team.
- Lower monitoring cost by using existing bus journeys.
- Better infrastructure planning using repeated, geo-tagged observations.
- Potential environmental benefit through smoother road conditions and traffic flow.
- Greater accountability through a traceable detection-to-resolution workflow.

5. Prototype Scope, Limitations & Future Development

Current Prototype Scope

The submitted browser prototype demonstrates the user-facing workflow and dashboard experience. It is designed to communicate how a production TransitIQ system could operate rather than claim that a city-scale sensing backend or trained field model is already deployed.

Future Development

- Connect real bus GPS and camera feeds.
- Train and validate the road-damage model on representative Indian road conditions.
- Deploy a production FastAPI/PostGIS backend and secure APIs.
- Integrate with municipal ticketing and department workflows.
- Add offline buffering and robust synchronization for low-connectivity routes.
- Validate detection accuracy, latency and response-time improvements through a controlled pilot.

Project Links

GitHub Repository: <https://github.com/akarsha57/TransitIQ>

References / Technology Documentation

- YOLOv8 — real-time object detection framework used in the proposed edge-AI architecture.
- FastAPI — proposed Python backend framework.
- PostGIS — proposed spatial database extension for location-based queries.
- Leaflet / OpenStreetMap — map and geographic visualization layer used by the prototype.

Prepared for Smart India Hackathon 2026 · TransitIQ · Team #ArtisanX